

MATERIALS' DATA BASE UPDATION USING LISTS

Aim: To write and execute a Python program for updation of materials' data base using lists.

Interactive Development Environment (IDE) Used : Jupyter Notebook

Programming Language Used : Python

Python Concepts used:

A List is an ordered collection of elements enclosed within square brackets ([]). Lists are mutable, allowing elements to be added and deleted. In this experiment, three parallel lists store steel grades, steel types and applications. A user-defined display() function is reused to display the list before and after updating.

List Operation / Concept	Example Code	Description
Create a List	<code>steel_grade = ["AISI 1018", "AISI 1045", "AISI 1095"]</code>	Creates a list containing multiple steel grades.
Display Entire List	<code>print(steel_grade)</code>	Displays all the elements of the list.
Access an Element	<code>print(steel_grade[1])</code>	Displays the second element of the list using its index.
Traverse a List	<code>for i in range(len(steel_grade)): print(steel_grade[i])</code>	Displays each element of the list one by one using a for loop.
Find Length of List	<code>len(steel_grade)</code>	Returns the total number of elements in the list.
Add a New Element	<code>steel_grade.append("EN8")</code>	Adds a new steel grade to the end of the list.
Delete an Element	<code>steel_grade.pop(index)</code>	Removes the element at the specified index.
Find Position of an Element	<code>index = steel_grade.index("AISI 1045")</code>	Returns the index (position) of the specified element.
Check Element Exists	<code>if "EN8" in steel_grade:</code>	Checks whether the specified element is present in the list.
Modify an Element	<code>steel_grade[1] = "EN24"</code>	Replaces an existing element with a new value.
Insert an Element	<code>steel_grade.insert(1, "EN8")</code>	Inserts a new element at the specified position.
Remove by Value	<code>steel_grade.remove("AISI 1018")</code>	Removes the specified element from the list.
Sort a List	<code>steel_grade.sort()</code>	Arranges the list elements in ascending order.
Reverse a List	<code>steel_grade.reverse()</code>	Reverses the order of the list elements.
Clear a List	<code>steel_grade.clear()</code>	Removes all elements from the list.

Python Program:

```
# Experiment 4: Classification of Steel Grades Using Lists
def display(steel_grade, steel_type, applications):
    for i in range(len(steel_grade)):
        print(i+1, ". Steel Grade :", steel_grade[i], "Steel Type :",
              steel_type[i], "Application :", applications[i])

# Main Program
steel_grade = ["AISI 1018", "AISI 1045", "AISI 1095"]
steel_type = ["Low Carbon Steel", "Medium Carbon Steel", "High Carbon Steel"]
applications = ["Bolts and Shafts", "Gears", "Springs"]
# Display Existing Steel Details
display(steel_grade, steel_type, applications)
choice = int(input("\nEnter your choice (1-Add, 2-Delete): "))
if choice == 1:
    g = input("Enter Steel Grade : ")
    s = input("Enter Steel Type : ")
    a = input("Enter Application : ")
    steel_grade.append(g)
    steel_type.append(s)
    applications.append(a)
    display(steel_grade, steel_type, applications)
elif choice == 2:
    grade = input("Enter Steel Grade to Delete : ")
    if grade in steel_grade:
        index = steel_grade.index(grade)
        steel_grade.pop(index)
        steel_type.pop(index)
        applications.pop(index)
        display(steel_grade, steel_type, applications)
    else:
        print("\nSteel Grade Not Found.")
else:
    print("\nInvalid Choice.")
```

Sample Input and Output :

Result :

The Python program was written and executed successfully updation of materials' data base using lists.